



Sheet of PW N°6

Exercise 1

Write a C program allowing to

- Read the number of elements to store in an array of integers,
- Read the elements of the array,
- Calculate and display the sum of the even numbers in the array,
- Calculate and display the sum of the odd numbers in the array.

Exercise 2

Write a C program to convert a decimal number to a binary number using successive division by 2.

Exercise 3

Let Mag be an array containing the quantity of a product X in different stores. Write a C program allowing to

- Read the number of stores,
- Read the quantity of product X in each store,
- Read the quantity requested by the user,
- Display a distribution of the quantity requested by the user if the sum of the quantities is sufficient. Otherwise, display missing quantity.

Example

The number of stores is : 5

The quantity of product X in each store is :

Store	1	2	3	4	5
Quantity	14	25	17	38	10

Let's assume the user requests 50 units,

The distribution to be displayed is as follows:

14 from store1 + 25 from store2 +11 from store3.

Exercise 4

Implement a game that allows a player to guess (يخمن) a secret combination of numbers. The idea is that the computer generates a secret combination of n digits (n must be in the interval [1, 9]). Then, the player suggests (يقترح) combinations. For each guess, the computer provides clues:

- a) Correct digit in the right position,
- b) Correct digit but in the wrong position,
- c) Incorrect digit.

When every one of the n positions is correct, the player wins. If the player fails to find the secret combination within 10 attempts, the game is over.

Exercise 5

Write a C program allowing to

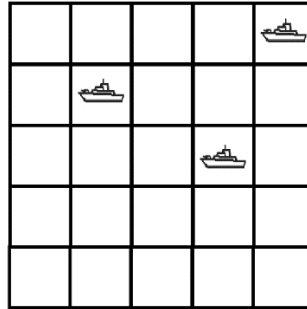
- Read the number of rows L and the number of columns C
- Read the (integers) elements of a matrix of size L x C,
- Calculate the sum of the elements in each column.



Exercise 6

Write a C program implementing a simplified version of the "Battleship" game.

Three hidden ships are randomly positioned on an $n \times n$ grid. Players guess coordinates to find them, each successful hit destroys a ship and increases the score. With only 10 attempts, players must locate all three ships to win.



يتم إخفاء ثلاث سفن بشكل عشوائي على شبكة بحجم $n \times n$. يحاول اللاعب تحديد موقعها من خلال تخمين إحداثياتها. كل ضربة ناجحة تدمر سفينة واحدة ويضيف نقطة واحدة إلى النتيجة. مع 10 محاولات فقط، يجب على اللاعب العثور على السفن الثلاث للفوز.

Knowing that, in the following C program :

- 1) The grid is represented by a matrix *grid*.
- 2) Ships are represented by value 1 in the matrix.

Complete this program by successively carrying out the following operations:

- Ask for coordinates of three ships
- Check if a ship is hit
- Display the result
- Count found ships

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

#define SIZE 5
#define MAX_ATTEMPTS 10

int i, x, y, row, column, attempts, ships_found;

int grid[SIZE][SIZE] = {0};

main() {
    attempts = 0;
    ships_found = 0;

    // Random ship initialization
    srand(time(NULL));

    // Placement of 3 ships
    for(i = 0; i < 3; i++) {
        do {
            x = rand() % SIZE;
            y = rand() % SIZE;
        } while(grid[x][y] == 1);
        grid[x][y] = 1;
    }
    printf("=== MATRIX NAVAL BATTLE ===\n");
    printf("Find the 3 ships in %d moves maximum!\n",
    MAX_ATTEMPTS);
}
```